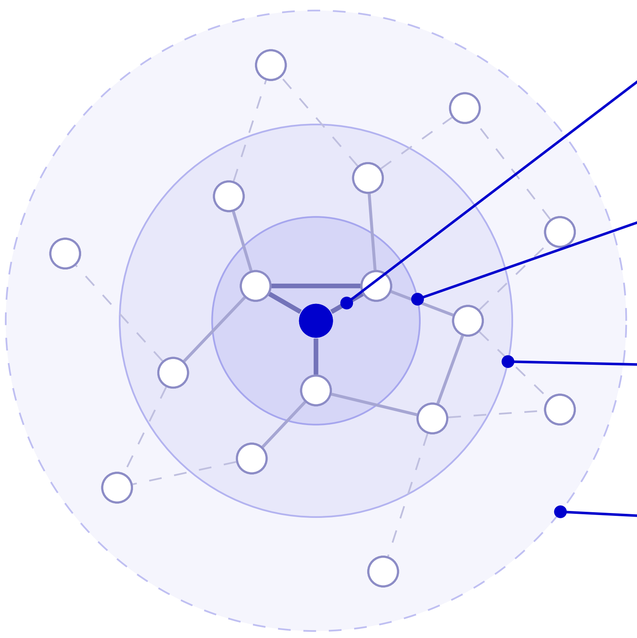
Beyond Co-Movement: Locality by Exposures

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Assets are close when they share risk exposures, not when they happen to move together.

Defining graph locality this way gives a market graph that is interpretable, stable and better for diversification.

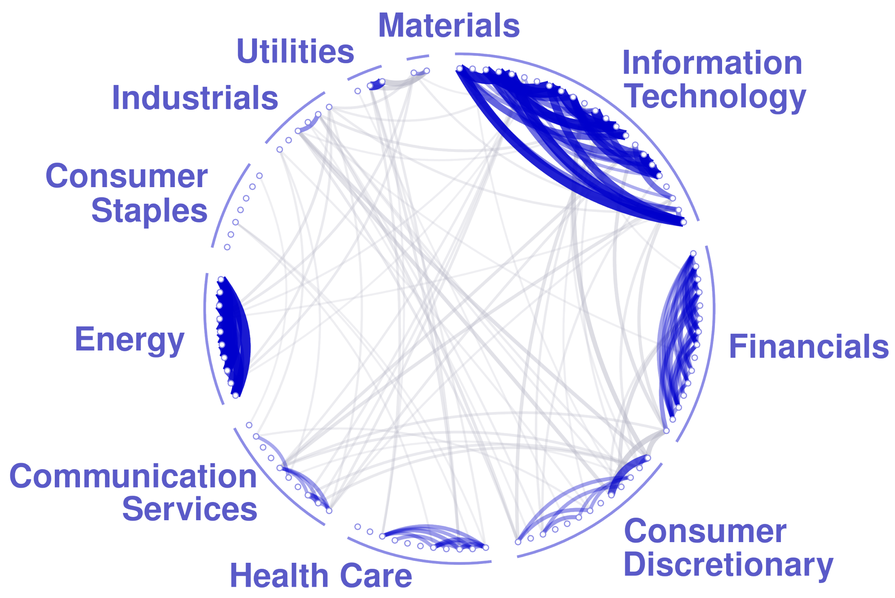


1 Edges encode closeness
Edge strength says how closely two assets are related

2 Small neighbourhoods
Capture idiosyncratic, firm-level behaviour

3 Wider neighbourhoods
Capture sector-wide dynamics

4 Widest neighbourhoods
Capture market-wide, systematic trends



Locality is a central modelling decision: it defines how vertices relate, and downstream tasks inherit any flaw in it.

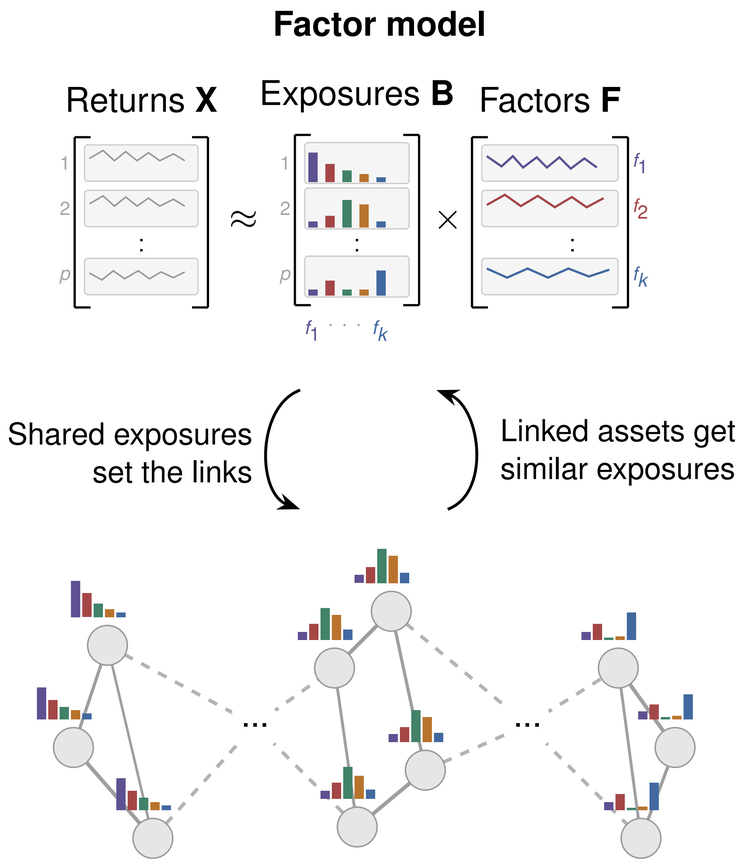
Traditionally, graph allocators diversify across neighbourhoods. Correlation-based graphs conflate persistent risk with noise and artefacts. As a result, every neighbourhood inherits that conflation.

Close = driven by the same risks — Our approach

1. Each asset carries its exposure profile
2. Links come from shared exposures
3. Groups reflect genuine structural similarity

MINGLE redefines locality by shared exposures

This ensures **mutual consistency**: the factor model sets the exposures that locality is measured on, and the graph smooths those exposures over neighbourhoods.



Graph topology W

Both the factor and graph domains are learnt jointly, in one objective given by

$$\min_{\mathbf{B},\mathbf{F},\mathbf{W}} \underbrace{\|\mathbf{X}-\mathbf{BF}\|_{\Omega}^{2}}_{\text{Factor decomposition}} + \underbrace{\lambda \sum_{i=1}^{p}\sum_{j=1}^{p} w_{ij}\,\|\mathbf{b}_i-\mathbf{b}_j\|_2^2}_{\text{Exposure–smoothness on graph}}$$

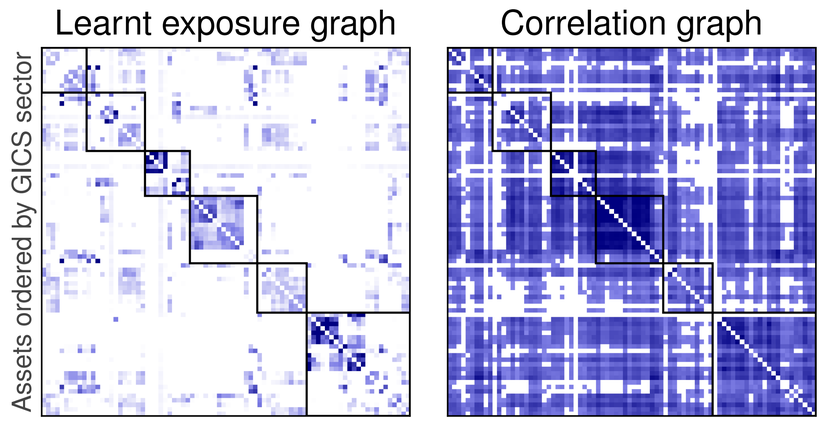
$$+ \underbrace{\alpha\|\mathbf{W}\|_1}_{\text{Graph sparsity}} \underbrace{-\beta\,\mathbf{1}^{\top}\log(\mathbf{W}\mathbf{1}+\varepsilon\mathbf{1})}_{\text{Degree barrier}} + \underbrace{\delta\|\operatorname{offdiag}(\mathbf{F}\Omega\mathbf{F}^{\top})\|_F^2}_{\text{Factor decorrelation}}$$

$$\text{s.t.} \quad \underbrace{\mathbf{B}^{\top}\mathbf{B}=\mathbf{I},}_{\text{PCA scale ambiguity}} \quad \underbrace{w_{ij}\geq 0\ \forall i,j,\ w_{ii}=0\ \forall i,\ \mathbf{W}^{\top}=\mathbf{W}}_{\text{Similarity graph topological constraints}}$$

The optimisation problem is solved with ADMM, and is re-estimated monthly.

The learnt graph recovers sectors, unsupervised

Each square is one asset pair, ordered by GICS sector. Dark blocks on the diagonal mean links stay within sectors.

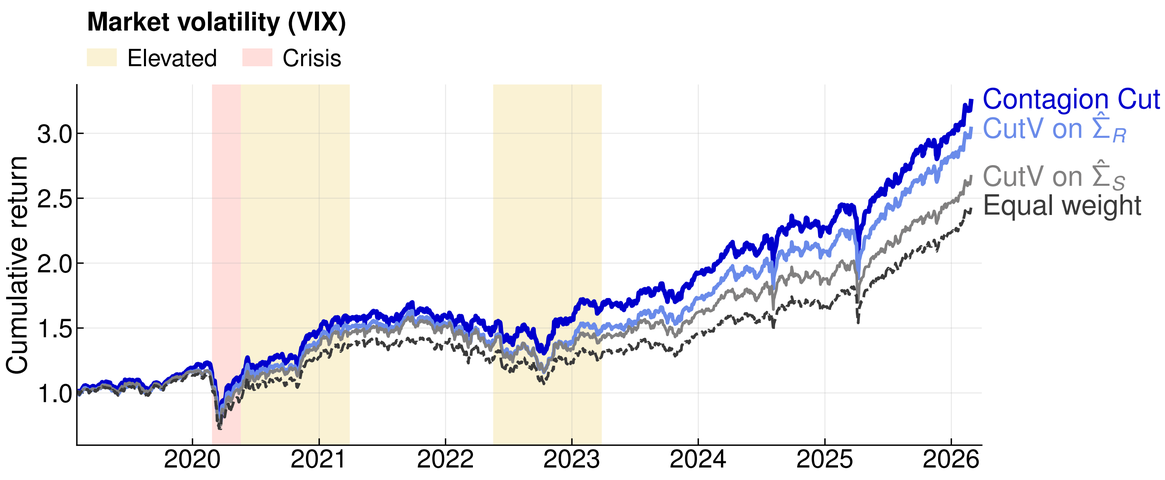


5.3× within- vs between-sector link strength (correlation: 2.4×)

0.77 NMI: communities persist across rebalancings

∼100× lower condition number than the sample covariance

Diversification improves as a consequence



The more of the learnt graph the allocator uses, the better it performs: CutV on $\hat{\Sigma}_R$ uses it to set weights *between* clusters; Contagion Cut also uses it *within* clusters.

Sharpe **1.07 vs 0.86** without costs, and **0.93 vs 0.80** at 50 bps (Jan 2019 – Mar 2026).

Each term matters

Dropping the low-rank or decorrelation term lowers Sharpe at every noise level ($p < 0.001$, 100 seeds). Joint beats sequential at low noise.

Limitations

Gains concentrate in Calm and Elevated regimes; long-only Sharpe stays negative in Crisis.

Setup

300 equities (S&P 500, Nikkei 225, STOXX 600); 2-year lookback; monthly rebalancing; $k = 6$ factors.

Paper

Code

IMPERIAL